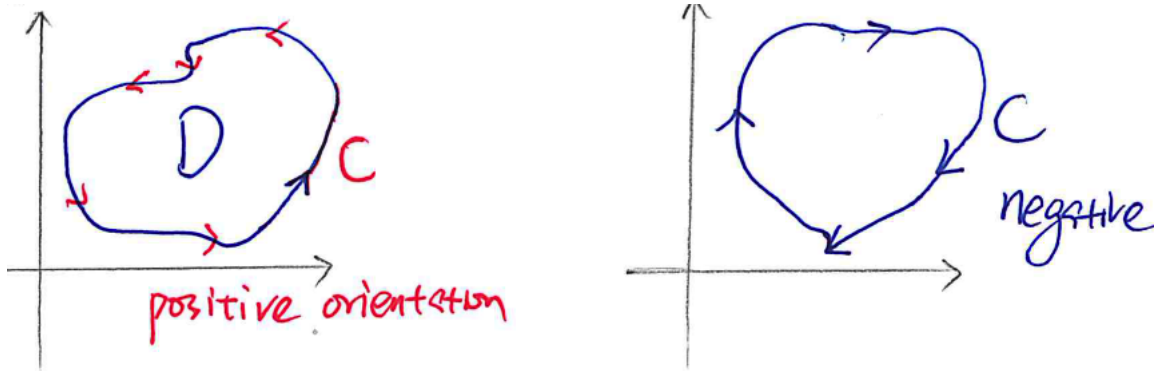


§4.4 Green's Theorem

Let $D \subset \mathbb{R}^2$ be a region bounded by a closed curve C . That is $\partial D = C$.

A **positive orientation** of C is the **counterclockwise** direction of C .

Suppose C is piecewise-smooth defined by the vector function $\vec{r}(t)$, $a \leq t \leq b$.



Theorem. Green's Theorem.

If $P(x, y)$ and $Q(x, y)$ have continuous partial derivatives, then

$$\int_C Pdx + Qdy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Recall that if $\vec{F} = P\vec{i} + Q\vec{j}$, then

$$\int_C \vec{F} \cdot \vec{r} = \int_C Pdx + Qdy.$$

Green's Theorem relates line integral and double integral.

If C is a closed curve, then we sometimes denote the line integral as

$$\oint_C Pdx + Qdy$$

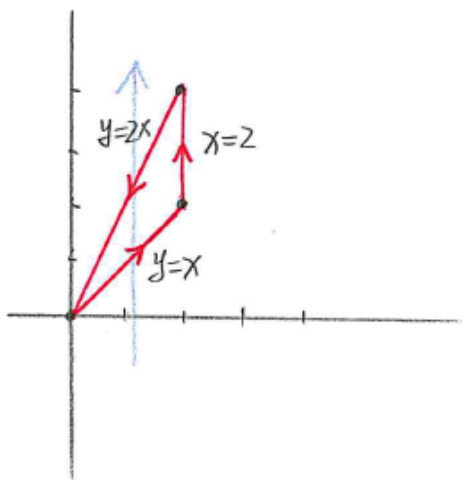
Green's Theorem sometimes is written as

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \int_{\partial D} Pdx + Qdy$$

First, we can use the double integral to compute the line integral by Green's Theorem.

If $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$, then by Green's Theorem $\int_C \vec{F} \cdot \vec{r} = 0$ for every closed path C in D . Then, $\vec{F}$ is conservative in D .

Example 1. Evaluate $\int_C xy^2 dx + 2x^2 y dy$ where C is the triangle from $(0,0)$ to $(2,2)$ to $(2,4)$ to $(0,0)$.



$$\begin{aligned}
 & \int_C xy^2 dx + 2x^2 y dy \\
 &= \iint_D (4xy - 2xy) dA \\
 &= \iint_D 2xy dA \\
 &= \int_0^2 \int_x^{2x} 2xy dy dx \\
 &= \int_0^2 xy^2 \Big|_x^{2x} dx \\
 &= \int_0^2 4x^3 - x^3 dx \\
 &= \int_0^2 3x^3 dx = \frac{3}{4} x^4 \Big|_0^2 = 12
 \end{aligned}$$

Example 2. Evaluate $\int_C (2y + e^{\cos x})dx + (6x - \ln(y^6 + y^2))dy$ where C is the circle $x^2 + y^2 = 9$.

$$\int_C P dx + Q dy$$

$$= \iint_D \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} dA$$

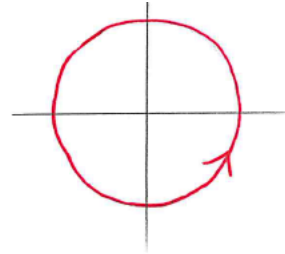
$$= \iint_D 5 - 2 dA$$

$$= \int_0^{2\pi} \int_0^3 4 \cdot r dr d\theta$$

$$= \int_0^{2\pi} d\theta \int_0^3 4r dr$$

$$= 2\pi (2r^2|_0^3)$$

$$= 36\pi.$$



polar system

$$x = r \cos \theta$$

$$y = r \sin \theta$$

Circle is

$$r = 3$$

$$0 \leq \theta \leq 2\pi$$

Sometimes it is easier to evaluate the line integral $\int_C Pdx + Qdy$, then we can use Green's Theorem to compute the double integral $\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$.

1. For example, if $P = Q = 0$ on the curve C , then from Green's Theorem

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \int_C Pdx + Qdy = 0.$$

We don't even need to know the value of P and Q on D .

2. **Application: Compute the area of D .**

(We know that Area of $D = \iint_D 1dA$)

In order to use Green's Theorem, we need

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1.$$

Some easy possibilities:

$$\begin{cases} P(x, y) = 0 \\ Q(x, y) = x \end{cases} \quad \begin{cases} P(x, y) = -y \\ Q(x, y) = 0 \end{cases} \quad \begin{cases} P(x, y) = -\frac{1}{2}y \\ Q(x, y) = \frac{1}{2}x \end{cases}$$

Green's Theorem implies:

Theorem.

$$\text{Area} = \oint_C xdy = - \oint_C ydx = \frac{1}{2} \oint_C xdy - ydx$$

Example 3. Find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

$$\text{Area} = \frac{1}{2} \int_C x dy - y dx$$

$$= \frac{1}{2} \int_0^{2\pi} a \cos t b \cos t dt - b \sin t a (-\sin t) dt$$

$$= \frac{1}{2} \int_0^{2\pi} ab \cos^2 t + ab \sin^2 t dt$$

$$= \frac{1}{2} \int_0^{2\pi} ab dt$$

$$= \frac{1}{2} ab 2\pi$$

$$= ab\pi$$

parametric equation:

$$x = a \cos t$$

$$y = b \sin t$$

$$0 \leq t \leq 2\pi$$